



Community-Acquired Pneumonia

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Community-acquired pneumonia is lung infection that develops in people who are not patients in a hospital, usually in people with a normal (competent) immune system, or in those who are immunocompromised, but pneumonia in people who are immunocompromised is discussed separately.

- Many bacteria, viruses, and fungi can cause pneumonia.
- The most common symptom of pneumonia is a cough that produces sputum, but chest pain, chills, fever, and shortness of breath are also common.
- Doctors diagnose community-acquired pneumonia by listening to the lungs with a stethoscope and by reading x-rays or computed tomography (CT) scans of the chest.
- Antibiotic, antiviral, or antifungal medications are used depending on which organism doctors believe has caused the pneumonia.

(See also [Overview of Pneumonia](#).)

Causes of Community-Acquired Pneumonia

Many organisms cause community-acquired pneumonia, including bacteria, viruses, fungi, and parasites. In most cases, the specific microorganism causing the pneumonia is not identified. However, doctors can usually predict which microorganisms are most likely to be causing the pneumonia based on the person's age and other factors, such as whether the person also has other diseases.

Community-acquired pneumonia can develop in people whose immune systems are normal (also called competent) or in people whose immune systems are impaired (also called immunocompromised people). However, when doctors use the term community-acquired pneumonia, they are generally referring to people whose immune systems are normal. Pneumonia in immunocompromised people is discussed separately (see [Pneumonia in Immunocompromised People](#)).

"Walking pneumonia" is a nonmedical term used to describe a mild case of community-acquired pneumonia that does not require bedrest or hospitalization. Some people even feel well enough to go to work and participate in other daily activities.

Bacterial causes of pneumonia

The most common bacterial causes of community-acquired pneumonia are

- *Streptococcus pneumoniae*
- *Haemophilus influenzae*
- *Chlamydophila pneumoniae*
- *Mycoplasma pneumoniae*

Streptococcus pneumoniae (pneumococcus) causes about many hospitalizations in the United States each year. There are over 90 types of pneumococci, but most serious disease is caused by only a small number of types. Pneumococcal pneumonia can be very severe, particularly in young children and older adults.

Haemophilus influenzae pneumonia may occur in adults but is more common among children. However, childhood infection has become much less common since children have been routinely vaccinated against *H. influenzae*. *H. influenzae* pneumonia is more common among adults who have underlying chronic lung disorders such as chronic obstructive pulmonary disease (COPD) and bronchiectasis.

Chlamydophila pneumoniae is the second most common cause of lung infections in healthy people aged 5 to 35 years. *C. pneumoniae* is commonly responsible for outbreaks of respiratory infection within families, in college dormitories, and in military training camps. It causes a pneumonia that is rarely severe and infrequently requires hospitalization. *Chlamydia psittaci* pneumonia (psittacosis) is a rare infection caused by a different strain of chlamydia and occurs in people who own or are often exposed to birds.

Mycoplasma pneumoniae causes infection very similar to that caused by *C. pneumoniae*. *M. pneumoniae* pneumonia is more common among older children and adults younger than 40, especially those living in crowded environments, such as schools, college dormitories, and military barracks. Although the illness is rarely severe, symptoms can last for weeks or even months.

Legionella pneumophila causes pneumonia and flu-like symptoms sometimes called Legionnaires' disease. It accounts for about 1 to 8% of all pneumonias. *Legionella* bacteria live in water, and outbreaks have occurred primarily in hotels and hospitals when the organism has spread through the air conditioning systems or water supplies, such as showers. No cases have been identified in which one person directly infected another.

Staphylococcus aureus causes pneumonia that is resistant to some types of antibiotics. This bacteria is known as community-acquired methicillin-resistant *Staphylococcus aureus* (CA-MRSA), and it can cause severe pneumonia, primarily in young adults. Since the year 2000, the number of cases of community-acquired pneumonia caused by *S. aureus* has increased, but the infection is still uncommon.

Pseudomonas aeruginosa is an especially common cause of pneumonia in people with cystic fibrosis and other lung diseases and in those with an impaired immune system.

Viral causes of pneumonia

Common viruses that can cause community-acquired pneumonia include

- SARS-CoV-2, the virus that causes [COVID-19](#)
- [Respiratory syncytial virus](#) (RSV)
- Adenoviruses
- [Influenza](#) viruses
- [Metapneumovirus](#)
- Parainfluenza viruses
- [Epstein-Barr virus \(rarely\)](#).
- Coxsackievirus (rarely)

The virus that causes [chickenpox](#) can also cause a lung infection. [Hantavirus](#) and [severe acute respiratory syndrome](#) (SARS) are also viral causes of pneumonia. Sometimes a bacterial pneumonia can occur after a lung infection caused by a virus.

Fungal causes of pneumonia

Fungal causes of community-acquired pneumonia include

- *Histoplasma capsulatum* ([histoplasmosis](#))
- *Coccidioides immitis* ([coccidioidomycosis](#))

Less common fungal causes of pneumonia include *Blastomyces dermatitidis* ([blastomycosis](#)) and *Paracoccidioides braziliensis* ([paracoccidioidomycosis](#)). *Pneumocystis jirovecii* commonly causes pneumonia in people who have [human immunodeficiency virus \(HIV\) infection](#) or a weakened immune system ([immunocompromise](#)).

Parasites that cause pneumonia

Parasites that cause community-acquired pneumonia in people who live in higher income countries include

- *Toxocara canis* or *T. cati* ([toxocariasis](#))
- *Dirofilaria immitis* ([dirofilariasis](#))
- *Paragonimus westermani* ([paragonimiasis](#))

Psittacosis: An Unusual Type of Pneumonia

Psittacosis (parrot fever) is a rare pneumonia caused by *Chlamydia psittaci*, a bacterium present mainly in birds such as parrots, macaws, parakeets, and lovebirds. It is also present in other birds, such as pigeons, finches, chickens, and turkeys.

Usually, people are infected by inhaling dust from the feathers or the waste of infected birds. The organism also may be transmitted by a bite from an infected bird and, rarely, from person to person in cough droplets. Psittacosis mainly occurs in bird fanciers or in people who work in pet shops or on poultry farms.

About 1 to 3 weeks after being infected, a person develops a fever, chills, fatigue, and loss of appetite. A cough develops, which is initially dry but later brings up greenish sputum (thick or discolored mucus). Fever persists for 2 to 3 weeks and then slowly subsides. The disease may be mild or severe, depending on the person's age and the extent of lung tissue involved.

Blood tests to determine whether the person has a certain type of antibody, suggesting recent infection with *Chlamydia psittaci*, are the most reliable method for confirming the diagnosis, but doctors usually suspect the infection in people who have a history of exposure to birds.

Bird breeders and owners can protect themselves by avoiding dust from the feathers and cages of sick birds. Importers are required to treat susceptible birds with a 45-day course of tetracycline, which generally gets rid of the organism.

People with psittacosis are treated with tetracycline taken by mouth for at least 10 days. Recovery may take a long time, especially in severe cases. The death rate may reach 30% in people with severe untreated psittacosis, although with appropriate treatment most people recover.

Symptoms of Community-Acquired Pneumonia

Symptoms of community-acquired pneumonia include

- A general feeling of weakness (malaise)
- Cough
- Shortness of breath
- Fever
- Chills
- Chest pain

Cough typically produces sputum (thick or discolored mucus) in older children and adults, but the cough is dry in infants, young children, and older people. Shortness of breath usually is mild and occurs mainly during activity. Chest pain is typically worse when breathing in or coughing. Sometimes people have upper abdominal pain.

Symptoms vary at the extremes of age. Symptoms of pneumonia in infants and toddlers may include irritability and restlessness. Older adults may be confused or have a decreased level of consciousness. Older adults and young children may be unable to communicate chest pain and shortness of breath. Fever is common but may not occur in older adults.

Diagnosis of Community-Acquired Pneumonia

- Usually a chest x-ray or computed tomography (CT) scan of the chest

No matter what type of pneumonia is suspected, doctors listen to a person's chest with a stethoscope to make a diagnosis. Chest x-rays or CT scans are usually also done to confirm the diagnosis.

Doctors usually do not need to do additional tests to determine what organism is causing the pneumonia, because most organisms are effectively treated by standard treatment and identifying the organism does not make a difference in treatment success. However, if doctors do need to identify the organism (such as in people whose condition does not improve while taking initial antibiotics), they usually try to grow or identify parts of the organism from a specimen of sputum, blood, or urine. Even when such testing is done, the organism is identified less than half the time.

Sometimes the person's symptoms or risk factors may suggest the cause of the pneumonia. For example, a bird fancier may have [psittacosis](#). So doctors may ask detailed questions about a person's work, hobbies, or recent travel. Certain combinations of risk factors and symptoms may suggest *Legionella* infection. In people who have first had symptoms typical of [influenza](#) or of [chickenpox](#), most pneumonia that develops is actually a bacterial pneumonia that took root in the lungs inflamed by the virus. However, sometimes influenza and chickenpox cause pneumonia directly.

Treatment of Community-Acquired Pneumonia

- Antibiotics
- Sometimes antiviral or antifungal medications

Doctors evaluate many factors to determine whether people can be safely treated at home or whether they should be hospitalized because of high risk of complications. Some of the factors include the following:

- Age
- Whether another disorder, such as cancer or a liver, heart, or lung disorder, is also present
- Whether there are worrisome findings on physical examination or testing
- Whether people are able to care for themselves or have someone to help them

Antibiotics are started as soon as possible. Doctors may prescribe antibiotics based on the severity of the infection and the risk of complications (see table [How Is Community-Acquired Pneumonia Treated?](#)). People are also given fluids, medications to relieve fever and pain, and oxygen if needed.

Because the causative organism is difficult to identify, doctors choose antibiotics based on the organisms that are most likely to be causing pneumonia.

With antibiotic treatment, most people with bacterial pneumonia improve. In people who do not improve, doctors look for an alternative diagnosis, unusual organisms, resistance to the antibiotic used for treatment, infection with a second organism, spread of the infection beyond the lungs (for example, into the lining of the lungs [pleura]), or some other disorder (such as a problem with the immune system or a blocked [obstructed] airway) that is delaying recovery.

To treat influenza pneumonia, the antiviral medications oseltamivir, zanamivir, or baloxavir can be given. To treat chickenpox pneumonia, acyclovir is given. If a person with a presumed viral pneumonia is very sick or does not improve within a few days after beginning treatment, doctors may prescribe antibiotics in case bacteria have also infected the lungs.

To treat fungal pneumonia, antifungal medications may be given.

Sometimes doctors do follow-up chest x-rays about 6 weeks after treatment to make sure that any abnormal findings on chest x-ray have resolved. Follow-up may be more important in people who smoke and in older adults to ensure that the abnormal findings seen on chest x-ray represent pneumonia only rather than an underlying cancer with pneumonia.

TABLE		
How Is Community-Acquired Pneumonia Treated?		
Severity	Possible Medications	Comments
Mild pneumonia in otherwise healthy people with no risk factors*	Amoxicillin	Medications are taken by mouth.
	Azithromycin	
	Clarithromycin	People are treated at home.
	Doxycycline	
Mild pneumonia in people with risk factors*	Amoxicillin	Medications are usually taken by mouth.
	Amoxicillin/clavulanate	
	Azithromycin	Often, more than one medication is taken.
	Cefpodoxime	
	Cefuroxime	People are treated at home.
	Clarithromycin	
	Doxycycline	

Severity	Possible Medications	Comments
	Levofloxacin Moxifloxacin	
Moderate pneumonia <i>or</i> Pneumonia in people who cannot care for themselves	Azithromycin Cefotaxime Ceftriaxone Levofloxacin Moxifloxacin	Medications are usually given intravenously in a hospital. Often, more than one medication is given.
Severe pneumonia in people with many risk factors*	Some of the same medications used for moderate pneumonia Aztreonam Cefepime Ciprofloxacin Imipenem/cilastatin Meropenem Piperacillin/tazobactam	Medications are usually given intravenously in an intensive care unit. Usually, more than one medication is given.
* Risk factors include heart or lung disorders, cancer, alcohol use disorder, age older than 65 years, recent use of antibiotics, and a weakened immune system (for example, because of HIV, organ transplantation, or use of medications that suppress the immune system).		

Prognosis for Community-Acquired Pneumonia

Most people with community-acquired pneumonia recover. However, pneumonia can be fatal, most often in infants and in older adults. The death rate is higher in *Legionella* infections, possibly because people who develop the disease are less healthy even before they become sick.

Prevention of Community-Acquired Pneumonia

Stopping smoking is the best way to prevent pneumonia.

Some pneumonias can be prevented by vaccination. Unvaccinated people who are at high risk of severe pneumonia and who are in close contact with a person who has influenza can be given the antiviral medications oseltamivir or zanamivir to prevent influenza.

Vaccines to prevent pneumonia

Vaccines are available that offer partial protection against pneumonia caused by

- The bacterium [Streptococcus pneumoniae](#)
- [Haemophilus influenzae type b](#) (only in children)
- [Influenza](#)
- [Chickenpox](#) (varicella; only in children)
- [COVID-19](#)
- Respiratory syncytial virus (RSV)



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